

Sriram Kalathoor

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EDUCATION

Georgia Institute of Technology	July 2026
<i>Ph.D. Aerospace Engineering; Minor in Applied Mathematics; Minor in Higher Education</i>	
<i>Thesis: Large-to-small scale dynamics, modeling and simulation of flow, reaction, and thermal behavior.</i>	
<i>Advisor: Prof. J. C. Oefelein</i>	
Georgia Institute of Technology	July 2022
<i>M.S. Mathematics; Specialization in Analysis, Geometry & Topology, and Differential Equations</i>	
Georgia Institute of Technology	May 2022
<i>M.S. Mechanical Engineering; Specialization in Thermo-fluids; Minor in Physics</i>	
Indian Institute of Technology, Madras	May 2017
<i>M.S. Aerospace Engineering</i>	
<i>Thesis: LES of screech in an afterburner</i>	
<i>Advisor: Prof. S. R. Chakravarthy</i>	
SRM University	May 2013
<i>B.S. Mechanical Engineering</i>	
<i>Thesis: Simulations of particle-laden flows with applications to fluidization</i>	

RESEARCH EXPERIENCE

Expertise in mathematical modeling, algorithm development, simulation and assessment of broad classes of engineering problems under various applied scenarios

High Performance Computing Lab, Georgia Tech	2020-
<i>Research Assistant; Design, modeling and simulation of reacting flows.</i>	
- Designed and analyzed a fuel injector for a supersonic commercial aviation jet engine, including a study of droplet cloud behavior, evaporation and heat transfer rates - Assessed wall cooling mechanisms and air-bleeding profiles to ensure operational integrity of a commercial supersonic jet engine - Evaluated flow and emissions profiles, and optimized flow rates as well as engine geometry to minimize emissions and thermal loads - Performed high-fidelity CFD simulations of turbulent reacting flow	
Ben T Zinn Combustion Lab, Georgia Tech	2019-20
<i>Research Assistant; Theoretical analysis of combustion noise,</i>	
- Performed theoretical and numerical analysis on flame-turbulence interaction - Used CFD and FEA to assess combustion noise in a commercial jet engine - Optimized combustor flow and structural design to meet FAA regulatory norms on engine noise	
Computational Combustion Lab, Georgia Tech	2017-19
<i>Research Assistant; Simulations of complex flows</i>	
- Used simulations to evaluate and optimize ignition in high-pressure combustion engines - Performed coupled CFD/FEA simulations to analyze wall-flame interaction in a jet-in-cross-flow - Developed and implemented a two-level-set model to capture pressure and stress profiles in model submarines - Assessed hypersonic re-entry flow behavior using magnetohydrodynamic turbulence and coupled structural FEA	

Rarefied Gas Dynamics Lab, IIT Madras

2013-17

Research Associate; Simulations of acoustics

- Developed and implemented a multi-scale model for capturing screech (high-frequency instability) in a proprietary afterburner, including a coupled acoustic-structural analysis
- Used COMSOL to evaluate duct acoustics of upstream engine plumbing and analyze its effect on fuel/air mixing, combustion efficiency and structural integrity

Ben T Zinn Combustion Lab, Georgia Tech

2012-13

Undergraduate Research Assistant; Numerical study of the hydrodynamics of a non-reacting jet-in-crossflow

Thermodynamics & Combustion Engineering Lab, IIT Madras

2010-12

Undergraduate Research Assistant; Experimental & numerical study of droplet evaporation in isotropic turbulence

SKILLS AND INTERESTS

- Software: ANSYS, COMSOL, SolidWorks, PTC Creo, AutoCAD, Pointwise, ICEMCFD
- Fabrication & machining: Lathe (CNC), drilling, milling, welding, 3D printing, metrology

TEACHING EXPERIENCE

Expertise in teaching a range of introductory to advanced courses on thermo-fluid sciences and math methods

Instructor of Record, Aerospace Engineering, Georgia Tech

2020-26

Graduate student lecturer in the Aerospace Engineering undergraduate program for

- AE 3030 Aerodynamics: Spring 2024, 2025 2026
- AE 4040 CFD & FEA: Fall 2023, 2024, 2025; (Fall 2019 Co-Instructor)
- COE 2001 Statics: Summer 2022 (Co-Instructor)
- AE 4803 Accident Causation & System Safety: Fall 2020 (Co-Instructor)

Designed courses, exams, homework assignments, projects, etc., delivered lectures and mentored large classes (typically 75-100 students)

Guest Instructor, Mechanical Engineering, Georgia Tech

2022-23

- ME 2016 Computing Techniques: Fall 2022
- ME 3340 Fluid Mechanics: Spring 2023
- AE 6042 Introduction to CFD: Spring 2026, Spring 2026

PUBLICATIONS

1. Coherent structure analysis in a backward facing step, **Sriram Kalathoor**, Doğukan Karahan and Cansu Uzay Karahan, *Bulletin of the American Physical Society*, 2024
2. Chemical analysis of emission characteristics in bluff body stabilized turbulent premixed methane-air flames, Anant Girdhar, **Sriram Kalathoor** and Jechiel Jagoda, *Bulletin of the American Physical Society*, 2024
3. A numerical study of emission characteristics in bluff body stabilized turbulent premixed methane-air flames in confined environments, Anant Girdhar, **Sriram Kalathoor**, Jechiel Jagoda and Lakshmi Sankar, *Proc. ASME Turbo Expo 2024*, GT2024-126815, pp. V03AT04A075, London. DOI: [10.1115/GT2024-126815](https://doi.org/10.1115/GT2024-126815)
4. On the dynamics of instability, transition, and turbulence in the GTRE *Kaveri* afterburner, *MS Thesis*, **Sriram Kalathoor**, IIT Madras, 2018

5. Multi-scale computational simulation of combustion instability and transition in a model afterburner, **Sriram Kalathoor** and Satyanarayanan Chakravarthy, *Proc. ASME Turbo Expo 2017*, GT2017-63805, pp. V04AT04A049, Charlotte. DOI: 10.1115/GT2017-63805
6. Mechanical agitation of bed in a motor-driven two-phase fluidized bed particle seeder, Ashwin Kannan and **Sriram Kalathoor**, *Intl. J. Engg . Tr. Tech.*, **4(10)**, 2013, pp. 4383-4387
7. Solid Particle Seeding Using Cyclone Gas Flow for Optical Flow Visualization: Design Parameters to be Considered, **Sriram Kalathoor**, *Adv. Aero. Sci. Tech.*, **3(2)**, 2013, pp. 57-64

PUBLICATIONS (UNDER REVIEW)

Under-review and pre-print drafts provided on request

1. Interfacial roughness spectra and finite-depth salt-finger mixing at a two-layer thermohaline interface, **Sriram Kalathoor**, *Submitted to J. Phys. Ocean.*; Preprint available at [arXiv.2606.21131](#)
2. Interfacial spectral memory as a state variable for finite-depth salt-finger exchange, **Sriram Kalathoor**, *Submitted to Deep Sea Res. I*; Preprint available at [arXiv.2606.26045](#)
3. Near-field combustion-noise source dynamics in a reacting supersonic temporal mixing layer, **Sriram Kalathoor** and Joseph Oefelein, *Submitted to Comb. Th. Model.*; Preprint available at [arXiv.2606.15626](#)
4. Lagrangian coherent structures in a supersonic reacting temporal mixing layer, **Sriram Kalathoor** and Joseph Oefelein, *Submitted to Chaos*; Preprint available at [arXiv.2606.20352](#)
5. Filtering effects on entropy transport and entropy-production structure in a supersonic reacting shear layer, **Sriram Kalathoor** and Joseph Oefelein, *Submitted to J. Fl. Mech. Rapids*; Preprint available at [arXiv.2606.15399](#)
6. Event-level compression–chemistry coupling in a supersonic reacting temporal mixing layer, **Sriram Kalathoor** and Joseph Oefelein, *Submitted to Combust. Flame*; Preprint available at [arXiv.2606.15628](#)
7. Receptivity and biorthogonal decomposition in a reacting temporal mixing layer, **Sriram Kalathoor** and Joseph Oefelein, *Submitted to Th. Comp. Fl. Dyn.*; Preprint available at [arXiv.2606.20819](#)

PUBLICATIONS (IN-PREPARATION)

1. Temporal organization of heat release and kinematic activity in a turbulent mixing layer, **Sriram Kalathoor** and Joseph Oefelein, *In preparation (proofreading stage)*
2. Box-filtering effects on time-series diagnostics of a reacting temporal mixing layer, **Sriram Kalathoor** and Joseph Oefelein, *In preparation (proofreading stage)*
3. Intermittent collective low-frequency organization in a turbulent reacting mixing layer, **Sriram Kalathoor** and Joseph Oefelein, *In preparation (proofreading stage)*
4. Modal energetics and disturbance-energy partition in a turbulent shear layer, **Sriram Kalathoor** and Joseph Oefelein, *In preparation (proofreading stage)*
5. Topology, mixing, compressibility, and reaction physics in a supersonic temporal mixing layer, **Sriram Kalathoor** and Joseph Oefelein, *In preparation (proofreading stage)*
6. Interaction budgets and multiscale coupling in a turbulent shear layer, **Sriram Kalathoor** and Joseph Oefelein, *In preparation (proofreading stage)*
7. Box-filter robustness of modal interaction diagnostics in a supersonic mixing layer, **Sriram Kalathoor** and Joseph Oefelein, *In preparation (proofreading stage)*

8. Fractal geometry and multiscale connectivity of a turbulent temporal mixing-layer interface, **Sriram Kalathoor** and Joseph Oefelein, *In preparation (proofreading stage)*
9. Local wave–vortex interaction in a turbulent supersonic shear layer, **Sriram Kalathoor** and Joseph Oefelein, *In preparation (proofreading stage)*
10. Reactive-acoustic wave packet dynamics in a supersonic reacting shear layer, **Sriram Kalathoor** and Joseph Oefelein, *In preparation (proofreading stage)*
11. Interfacial spectral memory as a state variable for finite-depth salt-finger exchange, **Sriram Kalathoor** , *In preparation (writing stage)*

INVITED TALKS

1. **2024-2026** — Graduate student advisor, Yellow Jacket Space Program, GT Supersonics & GTXR
2. **2023-2026** — Graduate student supervisor for undergraduate research, High Performance Computing Lab
3. **2023** — Invited panel member for the Graduate School Research Board’s discussion series on creating open and collaborative research environments
4. **2022** — Invited panel member for an Office of Undergraduate Education discussion series on undergraduate research
5. **2022** — Invited guest lecturer at the IIT Madras Department of Aerospace Engineering, for lectures on adaptive mesh refinement, and spectral dynamics of turbulent flows

INSTITUTIONAL SERVICE

GTAE Brown Bag Seminar Moderator and Judge	2024-26
<i>Moderate the weekly Brown Bag student seminars where honors and new graduate students present their research</i>	
Student Volunteer, American Physical Society	2018
<i>Assisted with organizing the APS Annual Division of Fluid Dynamics Meet, Atlanta, GA</i>	
Group Leader; GT6000 Community, Mentorship & Support, Georgia Tech	2022-25
<i>Serve as a group and community engagement leader for GT6000 (Community, Mentorship & Support), an extended orientation program for incoming graduate students</i>	
Resident Assistant; Graduate Housing, Georgia Tech	2020-25
<i>Resident Assistant with Georgia Tech’s Graduate & Family Housing community</i>	
Georgia Tech EcoReps	2020-25
<i>Promoting and facilitating eco-friendly campus operations; also serve as the student head of the composting program across campus residence halls</i>	
Member, Graduate Student Emerging Leader Advisory Board, Georgia Tech	2023-24
<i>Serve on the Student Emerging Leaders Advisory Board (by nomination); meet/consult with campus administration and support resources to promote and improve student experience, and represent student voices and perspectives</i>	
Practice Coordinator, Yellow Jacket Archery Club	2019-24
<i>Coordinate practices, induction of new members, skill progression, coaching & training of 200 members in the Georgia Tech Archery Club</i>	
Hygiene Heroes Program Intern, University of California, Berkeley	2014-16
<i>Designed, piloted and analyzed an activity-based health curriculum for implementation in school systems world-wide</i>	

AWARDS

1. **2025** — AE Outstanding Graduate Student Instructor Award (AE 4040, Fall 2025), Georgia Tech Center for Teaching & Learning
2. **2024-25** — Course Instructor Honor Roll (AE 3030, Spring 2024 & Spring 2025) for top instructor feedback/ratings, Georgia Tech Center for Teaching & Learning
3. **2024** — AE Outstanding Graduate Teaching Assistant Award (AE 4040, Fall 2023), Georgia Tech Center for Teaching & Learning
4. **2023** — Community Developer of the Year Award, Georgia Tech Department of Housing
5. **2022** — CIRTL Associate Fellow, Georgia Tech Center for Teaching & Learning
4. **2021** — Invited (virtual) guest lecturer at the IIT Madras Department of Aerospace Engineering, for a lecture series on multiscale modeling approaches for capturing combustion instability and transition in jet propulsion devices

CERTIFICATIONS (CURRENT)

1. **USA Archery** Level 2 Archery Instructor and Level 2 National Judge
2. **American Red Cross** CPR/First Aid/AED/Concussion Certification
3. **QPR Institute** QPR/Suicide Prevention Gatekeeper